

NETC 280 – Lab 8 – Inventory and Asset List Creation

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Note: For this lab, the use of AI is not acceptable. All written work must be in your own words. You may use the internet to research device specifications and naming conventions. Do not copy and paste AI-generated text — this constitutes plagiarism and will result in a zero on the lab. Document any outside sources used.

Purpose:

This lab is the hands-on companion to the Week 14 lecture on Inventory and Asset List Creation. In the lecture you learned why organizations track assets, how records are structured, how assets are tagged, and how depreciation works. Now you will apply those concepts directly.

You will learn how to:

- Distinguish between asset types and correctly classify real-world items
- Design a standardized asset record with all required data fields
- Build a functional IT asset register in Excel using proper naming conventions
- Calculate straight-line depreciation and determine current book value
- Identify and correct common asset record problems such as ghost assets, bad naming, and missing ownership
- Reflect on how asset tracking connects to the security and compliance concepts from earlier in the course

Lab 8, Part 1: Asset Classification

The lecture covered six asset types: Fixed/Physical, IT, Intangible, Current/Consumable, Financial, and Infrastructure. It also explained that the same physical item can be classified differently depending on which organization owns it and how they use it.

1. For each item below, identify the asset type, state whether it is an asset or inventory for the organization described, and write a 1–2 sentence justification. Answer in your own words — do not simply repeat the lecture definitions.
 - A laptop computer used daily by an accountant at a CPA firm
 - 10 identical laptops sitting in sealed boxes in a Best Buy stockroom
 - A Microsoft 365 E3 license assigned to one employee
 - A box of printer paper in the supply closet of a law firm
 - The HVAC system in a company's server room
 - A Cisco ASA 5516-X firewall protecting a business network
2. The lecture used a laptop as an example of an item whose classification depends on context. In your own words, explain what the “context matters” principle means for asset classification. Give one original example (not used above) where the same physical item would be classified differently depending on which organization owns it. Answer in 3–5 complete sentences.

Part 2, Step 1: Set Up Your Spreadsheet

1. Open Microsoft Excel and create a new blank workbook.
2. Rename the first tab: Asset Register
3. Create the following column headers in Row 1. Bold all headers and apply a dark fill with white text.

Use these exact column headers, in this order:

- A – Asset ID (format: NETC-XXX, e.g. NETC-001)
- B – Asset Name (standardized name using the naming convention below)
- C – Asset Type (IT / Fixed / Intangible / Infrastructure / Consumable)
- D – Serial Number / License Key
- E – Location (Building and Room, or “Cloud” for hosted assets)
- F – Assigned To (full name and department)
- G – Acquisition Date
- H – Purchase Cost (\$)
- I – Useful Life (Years)
- J – Salvage Value (\$)
- K – Annual Depreciation (\$) — formula: $= (H2 - J2) / I2$
- L – Current Book Value (\$) — formula: $= H2 - (K2 * (YEAR(TODAY()) - YEAR(G2)))$
- M – Condition (New / Good / Fair / Poor / Decommissioned)
- N – Warranty Expiry
- O – Notes

Note: For columns K and L, enter the formulas exactly as shown above in Row 2, then copy them down for all rows. Excel will adjust the row references automatically.

Part 2, Step 2: Naming Convention

The lecture emphasized that asset names must follow a consistent naming convention. For Column B (Asset Name), use this format:

[Type]-[Dept]-[Year]-[Sequence] Example: PC-HR-2022-001 or SW-IT-2021-001 or LIC-SLS-2024-001

Type abbreviations: PC (desktop/laptop), SW (switch/network device), FW (firewall), LIC (software license), PRN (printer), UPS (battery backup), CAM (security camera).

Part 2, Step 3: Enter the Asset Data

Enter the following 8 assets into your register. Use the information provided and fill in any missing fields using your own judgment. Look up the serial number format or typical warranty period for each device type if needed.

1. Dell OptiPlex 7090 Desktop PC — purchased 03/15/2022 for \$1,249.00 — assigned to Sarah Kim, HR Dept
2. Cisco Catalyst 3650 24-Port Switch — purchased 07/01/2021 for \$3,800.00 — IT Dept, Server Room
3. Microsoft 365 Business Standard License (1 user) — purchased 01/01/2024 for \$150.00/yr — Marcus Webb, Sales (use 1 year useful life, \$0 salvage)
4. HP LaserJet Pro M404dn Printer — purchased 09/10/2020 for \$349.00 — Front Office, Room 101
5. Cisco ASA 5516-X Firewall — purchased 11/30/2019 for \$7,500.00 — IT Dept, Server Room
6. MacBook Pro 14" (M3 chip) — purchased 02/28/2024 for \$1,999.00 — Jamie Torres, Design Dept
7. APC Smart-UPS 1500VA Rack Mount — purchased 07/01/2021 for \$820.00 — IT Dept, Server Room
8. Verkada CD52 Security Camera (indoor) — purchased 05/01/2023 for \$499.00 — Facilities, Main Lobby

Part 2, Step 4: Format and Save

1. Apply alternating row shading for readability.
2. Freeze Row 1 so the header stays visible when scrolling.
3. Apply currency formatting to columns H, J, K, and L.
4. Apply date formatting to columns G and N.
5. Auto-fit all column widths.
6. Save the file as: LastName_FirstName_Lab8_AssetRegister.xlsx

Lab 8, Part 3: Depreciation Analysis

The lecture covered three depreciation methods: straight-line, declining balance, and units of production. Your Excel register uses straight-line. In this part you will analyze the results and answer questions that go beyond the formula.

1. Look at the Current Book Value column (L) in your completed register and answer the following:
 - a. Which asset currently has the lowest book value? What is it?
 - b. Which asset has depreciated the most in total dollar amount since purchase?
 - c. Are any assets showing a book value near \$0 or below? If so, which one(s)?
2. The lecture introduced the concept of “Replace vs. Refresh” as a practical use of depreciation data. Look at the Cisco ASA 5516-X Firewall in your register. Based on its current book value and the fact that it is a security-critical device, write a 3–5 sentence paragraph answering: Should this device be replaced, refreshed (upgraded), or kept as-is? Use the book value data from your register and reasoning from the lecture to support your recommendation.

3. You used straight-line depreciation for all 8 assets. However, the lecture explained that the declining balance method is often more appropriate for IT equipment because it loses value faster in early years. Identify one asset from your register where declining balance would more accurately reflect real-world value loss, and explain why in 2–4 sentences.

Lab 8, Part 4: Audit a Broken Asset Register

The lecture's Pitfalls slide identified three major asset tracking problems: ghost assets, bad data (inconsistent naming/missing fields), and no ownership assigned. In this part you are given a sample asset register that contains all three types of problems. Your job is to find them and document what is wrong.

Scenario: You have just been hired as a junior IT technician at a small manufacturing company. Your supervisor hands you their existing asset register and says: "We haven't touched this in two years — it's a mess. Find everything that's wrong with it before we do the physical audit next week."

Review the following asset records carefully:

Row	Asset Name	Serial #	Location	Assigned To	Purchase Date	Purchase Cost
1	Laptop	SN-48291	Office	Tom	2019	\$1200
2	PC-MFG-2019-001	DEL-7B29X-001	Floor 1, Room 104		04/01/2019	\$1,100.00
3	notebook	SN-48291	Warehouse	Luis Reyes, Ops	03/15/2022	\$989.00
4	SW-IT-2020-001	CSC-3750X-4422	Server Room	IT Dept	06/10/2020	\$4,200.00
5	Old Server	N/A	Server Room	IT Dept	2011	unknown
6	PRINTER-OFFICE	HP-LJ-9901	2nd Floor	Admin Staff	08/22/2021	\$320.00
7	PC-MFG-2019-001	DEL-7B29X-001	Floor 2, Room 208	Carla Voss, QA	04/01/2019	\$1,100.00
8	Security Cam	CAM-1190X	Lobby	Facilities	09/05/2018	\$450.00

Note: There are at least 6 distinct problems in this register. Look carefully at serial numbers, asset names, missing fields, duplicates, and assets that cannot still be in active use based on their age and description.

For each problem you find, document the following: the row(s) involved, the problem type (choose from: Ghost Asset, Bad/Inconsistent Naming, Missing Ownership, Duplicate Record, Missing/Invalid Data, or Other), what is specifically wrong, and how you would fix it. Find at least 5 problems.

Problem 1:

Problem 2:

Problem 3:

Problem 4:

Problem 5:

Problem 6 (if found):

Lab 8, Part 5: Reflection Questions

Answer each question after completing Parts 1–4. Each answer should be 3–5 complete sentences in your own words.

1. In Lab 4, you analyzed the Equifax breach using the NIST CSF. One of Equifax's critical failures was that their IT scans failed to identify all systems running the vulnerable Struts software — in other words, they had an incomplete asset inventory. Based on what you did in this lab, describe how a well-maintained IT asset register could have helped prevent or limit the Equifax breach. Be specific about which fields in your register would have been most useful.
2. The lecture's Key Risk slide warned that the most dangerous mistake in asset management is treating it as a one-time task. After building your own register in this lab, describe one specific event or workflow from real IT operations that would trigger an update to your register. Explain what field(s) would change, who would be responsible for making the update, and what could go wrong if the update was not made.
3. Imagine you are setting up asset tracking for a warehouse that holds 3,000 pieces of industrial equipment spread across four buildings. Based on the tagging methods covered in the lecture — barcode, QR code, RFID, and engraving — which method would you recommend, and why? What are the main trade-offs of your choice?

Turn in:

Submit a single .zip file that contains the following two files:

- LastName_FirstName_Lab8_AssetRegister.xlsx — your completed Excel asset register from Part 2
- LastName_FirstName_Lab8_Questions.docx — a single Word document containing your answers to all questions in Parts 1, 3, 4, and 5, clearly labeled by part and question number

Note: Label each answer clearly (e.g., “Part 1 – Question 1”) so your instructor can find each section without searching. Include a list of any outside sources you used to look up device specifications or naming conventions.